

# Properties of Regular Languages

For regular languages  $L_1$  and  $L_2$   
we will prove that:

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

Reversal:  $L_1^R$

Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$

Are regular  
Languages

A useful transformation: use one accept state

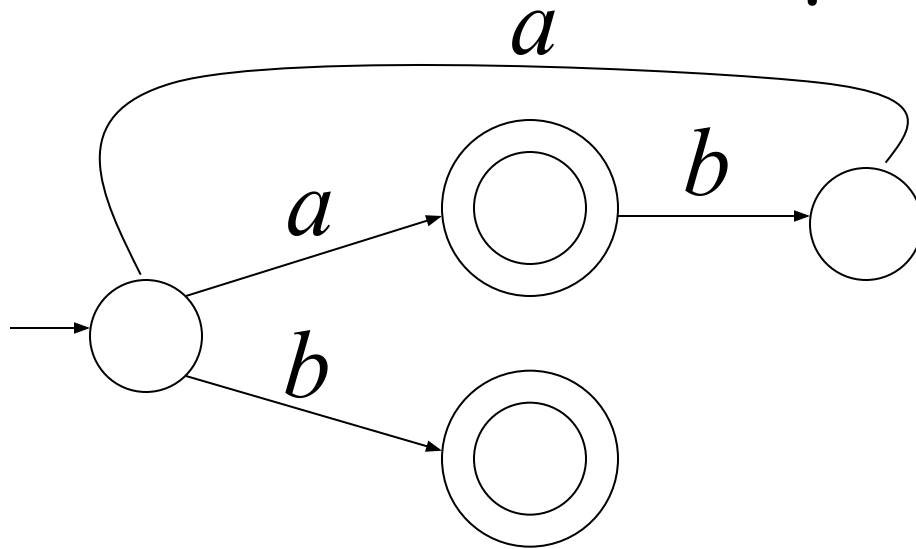
Any NFA can be converted

to an equivalent NFA

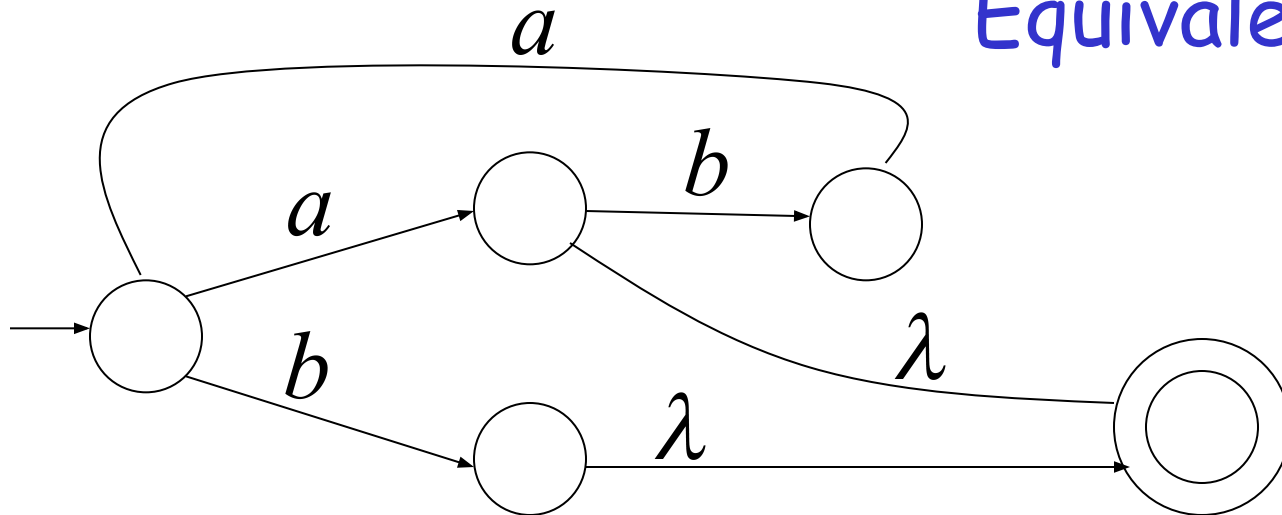
with a single final state

# Example

NFA

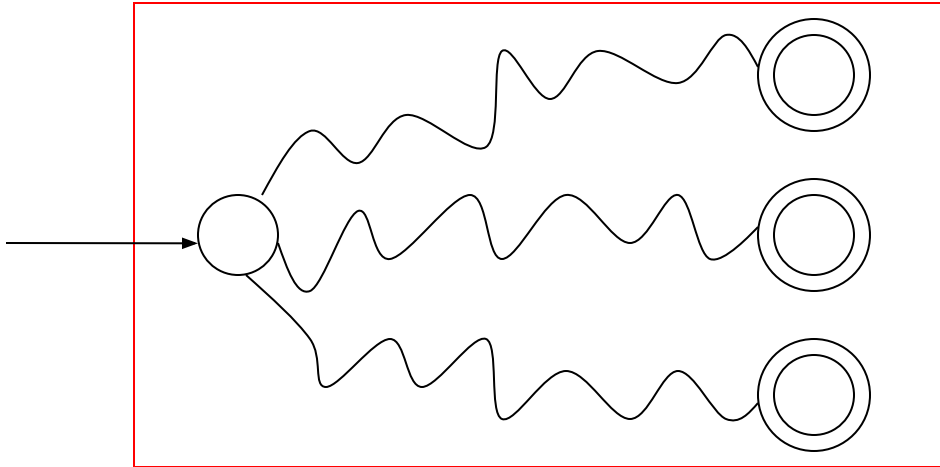


Equivalent NFA

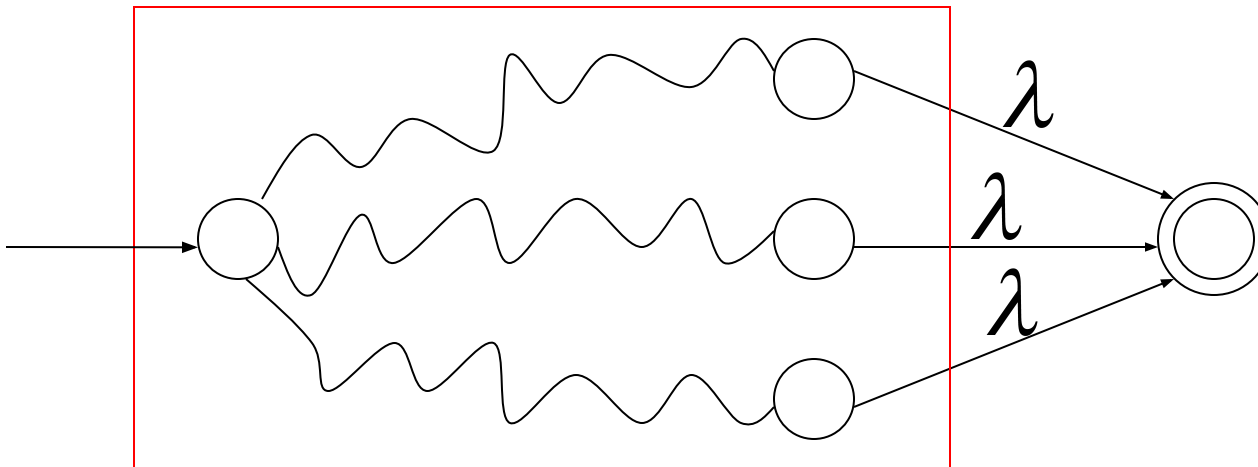


# In General

NFA



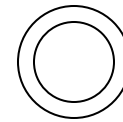
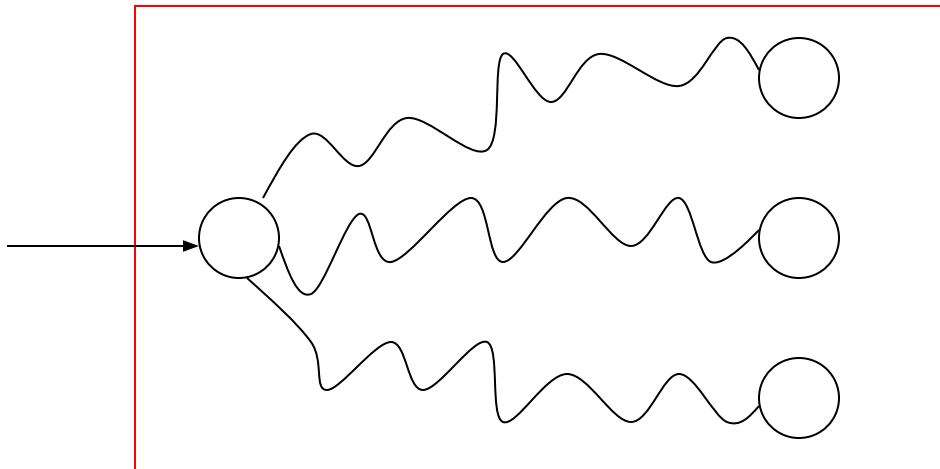
Equivalent NFA



Single  
final state

# Extreme Case

NFA without final state



Add a final state  
Without transitions

We say: Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

Reversal:  $L_1^R$

Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$

# Let us take two languages

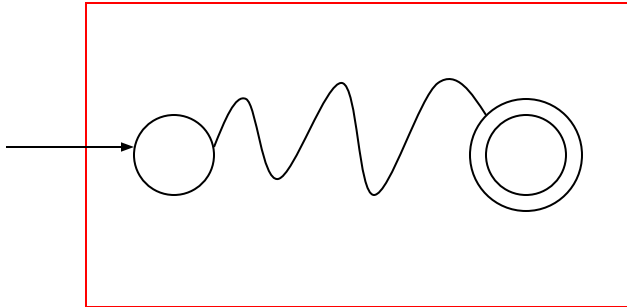
Regular language  $L_1$

Regular language  $L_2$

$$L(M_1) = L_1$$

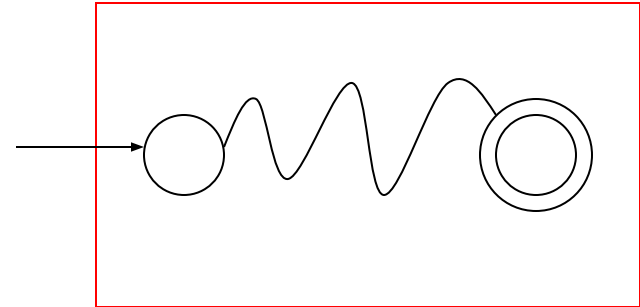
$$L(M_2) = L_2$$

NFA  $M_1$



Single final state

NFA  $M_2$

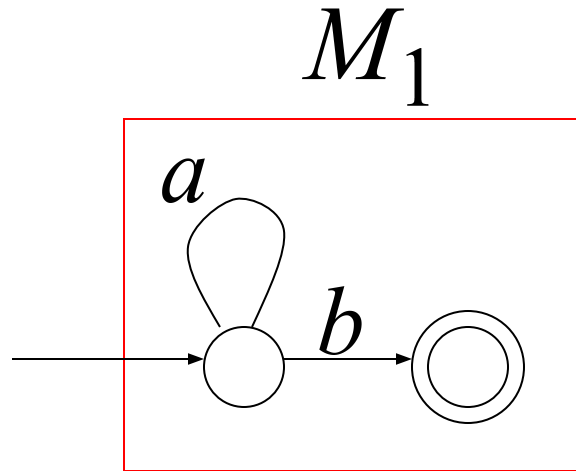


Single final state

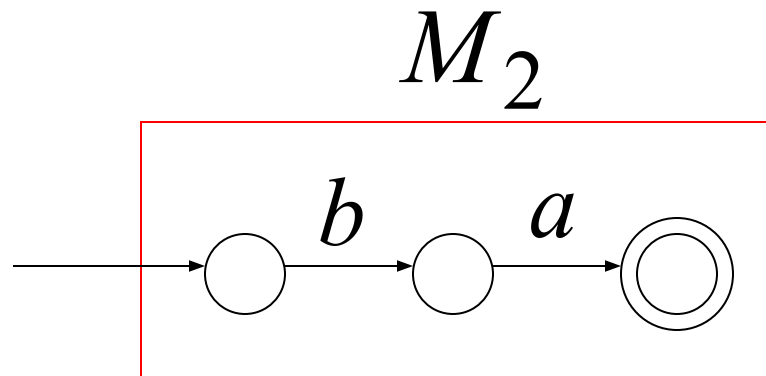


# Example

$$L_1 = \{a^n b \mid n \geq 0\}$$



$$L_2 = \{ba\}$$



# Show Regular languages are **closed under**

$$\text{Union: } L_1 \cup L_2$$

$$\text{Concatenation: } L_1 L_2$$

$$\text{Star: } L_1^*$$

$$\text{Reversal: } L_1^R$$

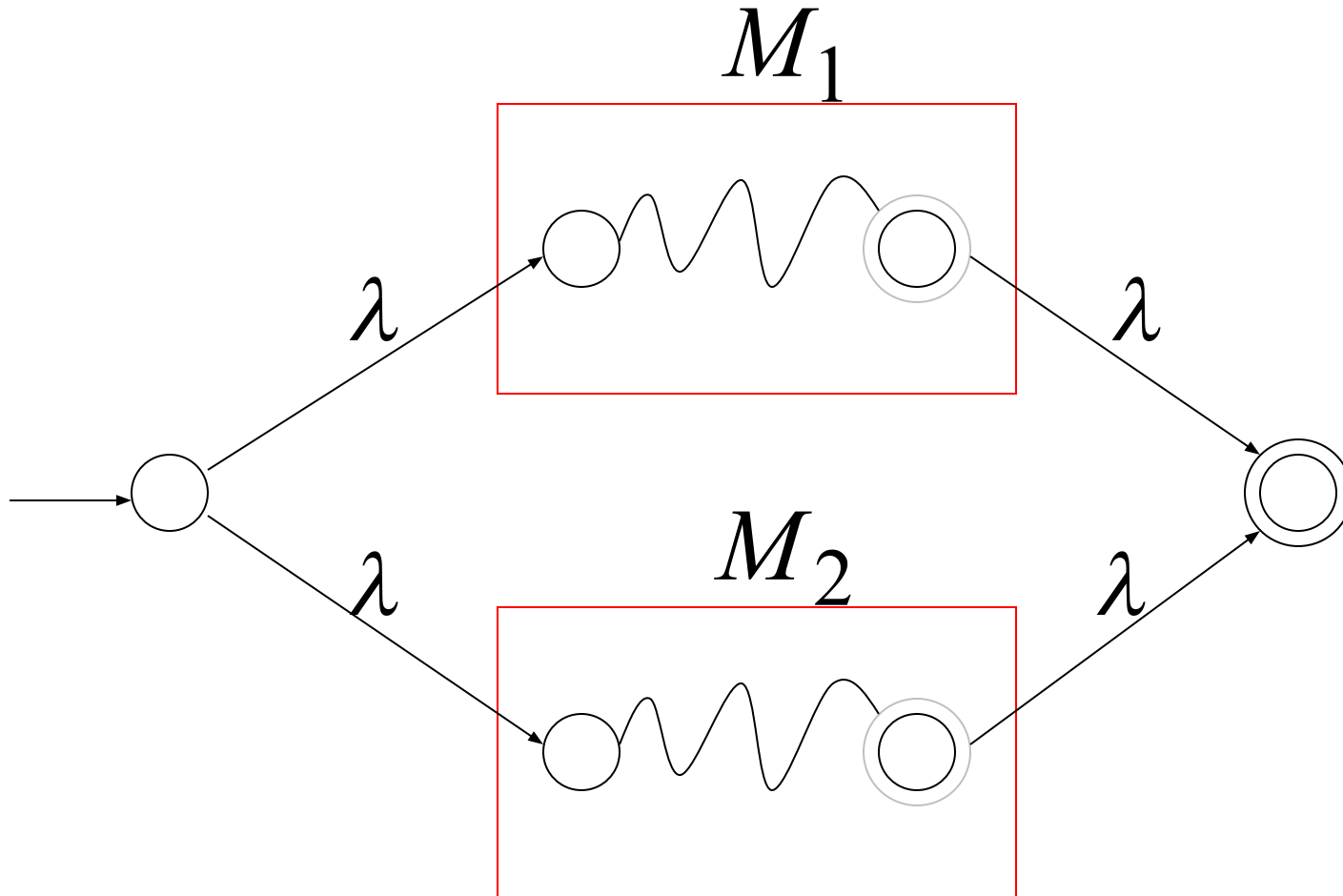
$$\text{Complement: } \overline{L_1}$$

$$\text{Intersection: } L_1 \cap L_2$$

HOW?

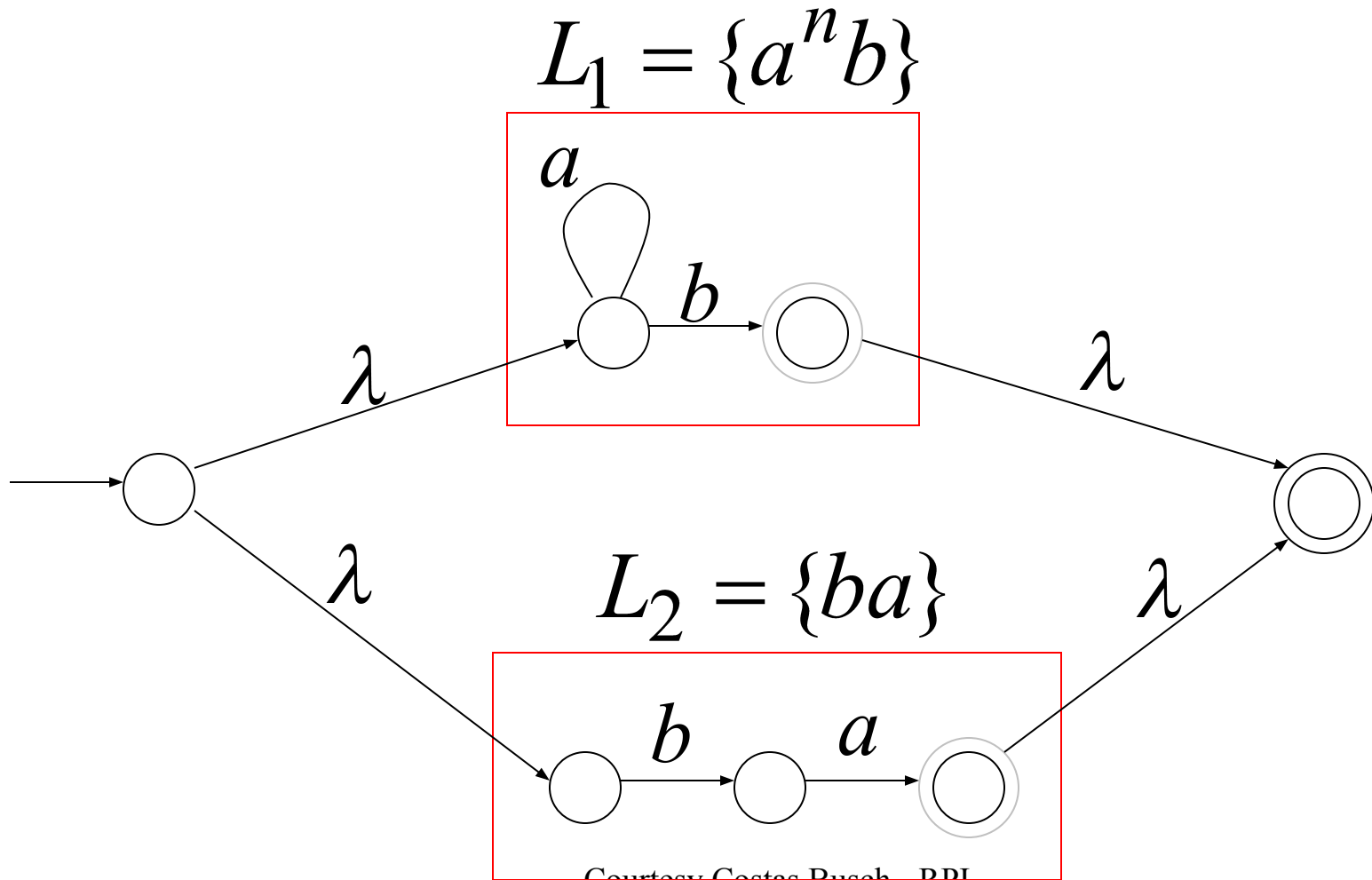
# Union

NFA for  $L_1 \cup L_2$



# Example

NFA for  $L_1 \cup L_2 = \{a^n b\} \cup \{ba\}$



Show Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

Reversal:  $L_1^R$

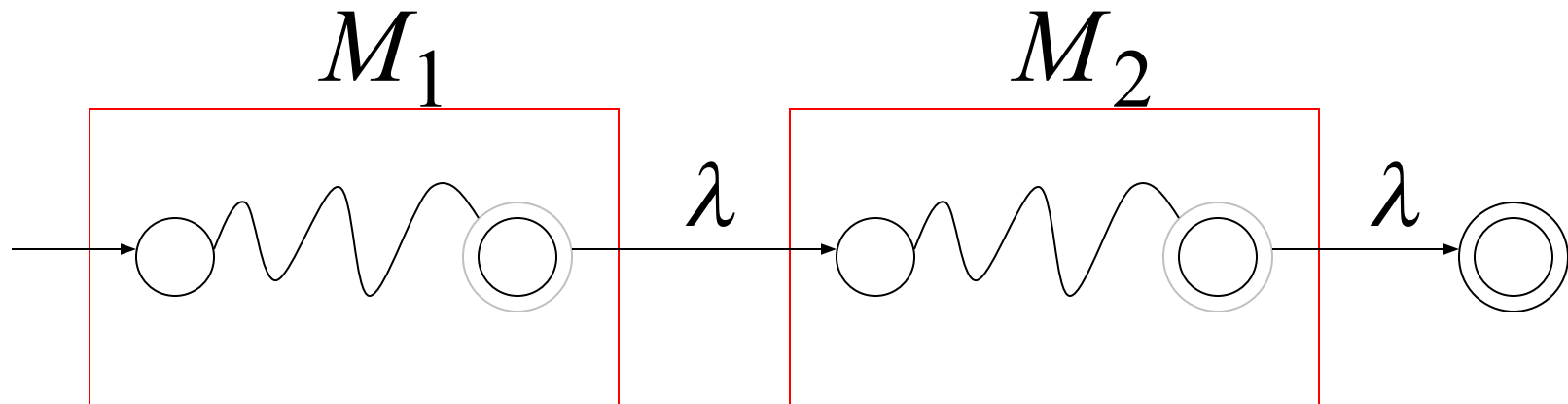
Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$

HOW?

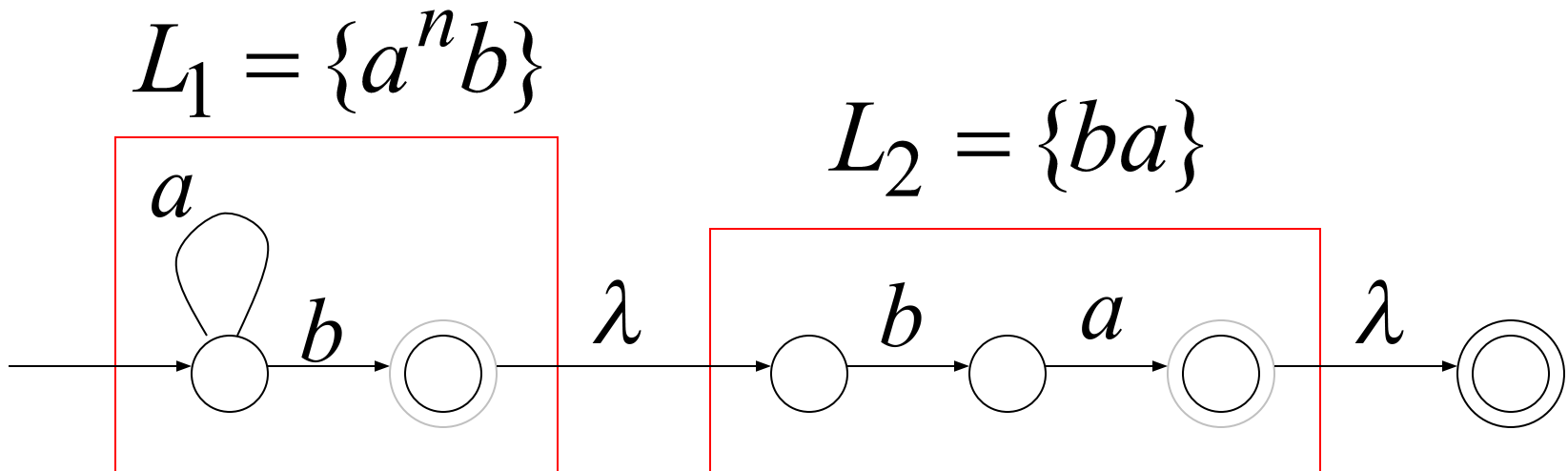
# Concatenation

NFA for  $L_1L_2$



# Example

NFA for  $L_1L_2 = \{a^n b\} \{ba\} = \{a^n bba\}$



# Show Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

HOW?

Reversal:  $L_1^R$

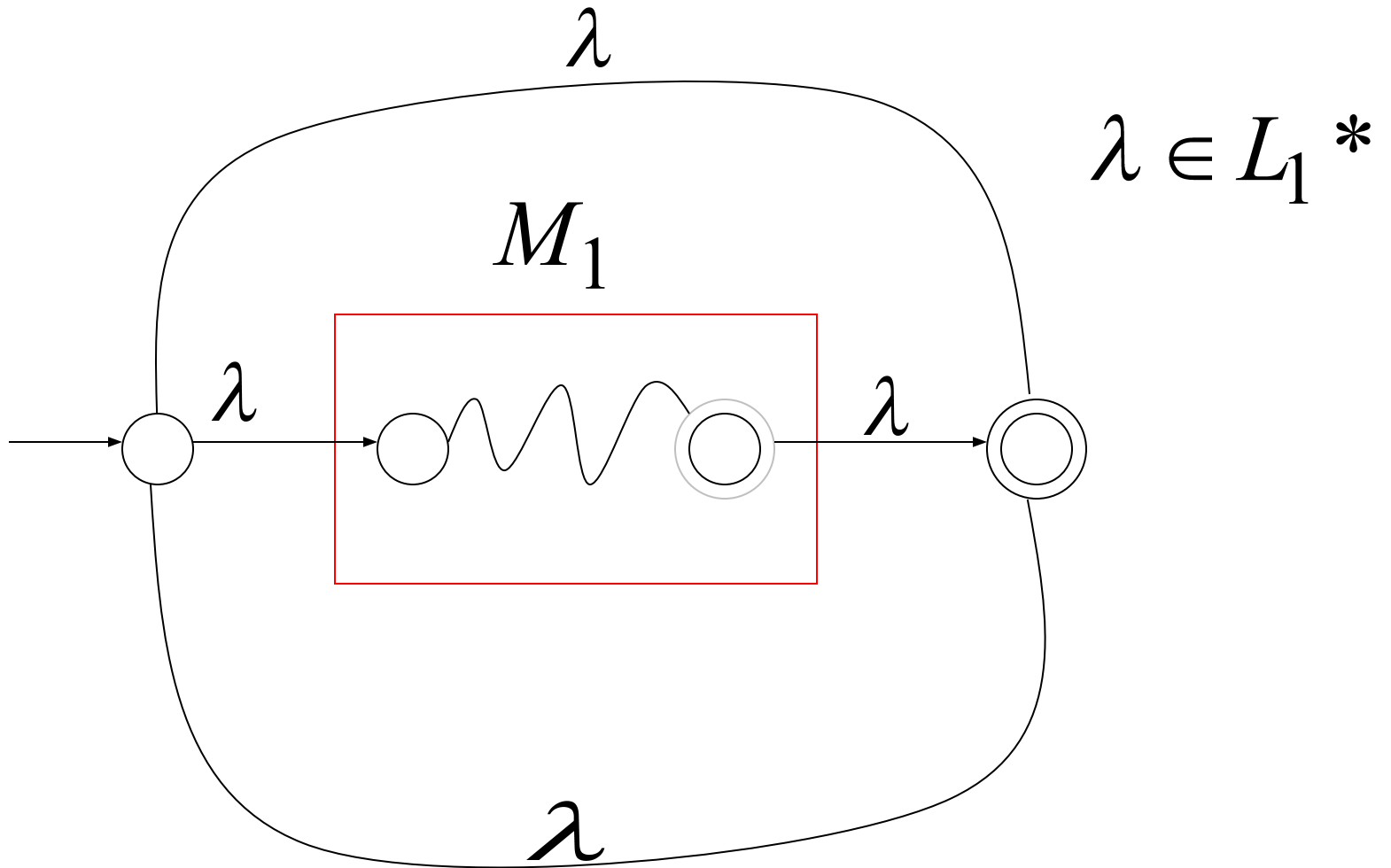
Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$



# Star Operation

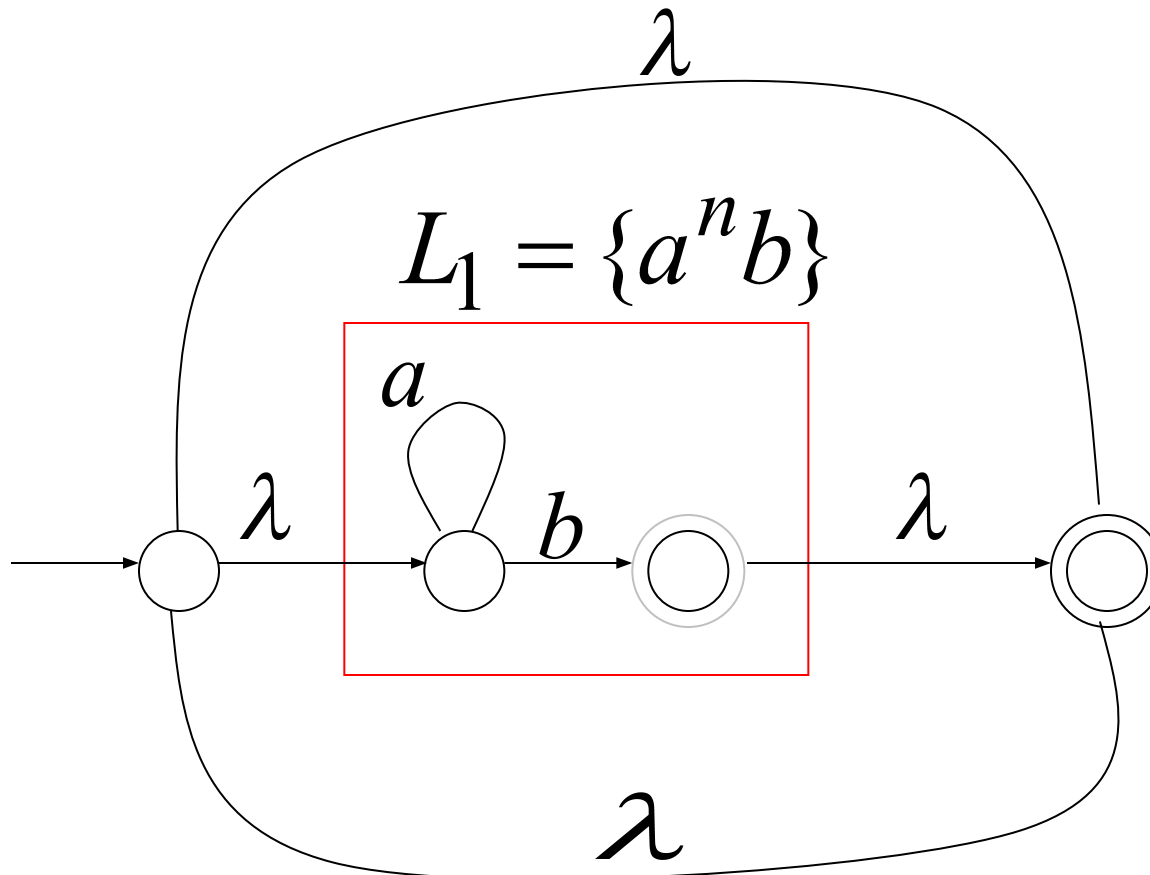
NFA for  $L_1^*$



# Example

NFA for  $L_1^* = \{a^n b\}^*$

$$w = w_1 w_2 \dots w_k$$
$$w_i \in L_1$$



Show Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

Reversal:  $L_1^R$

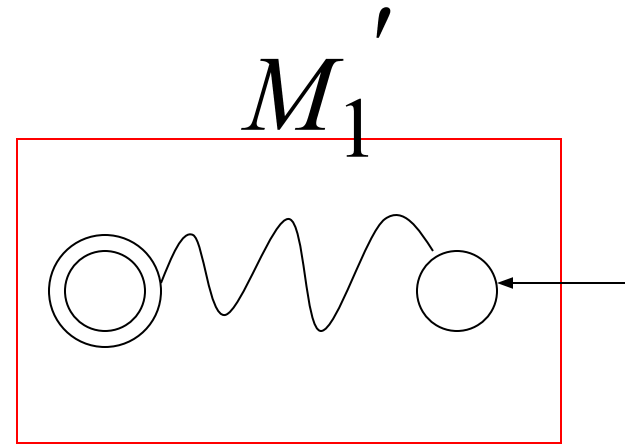
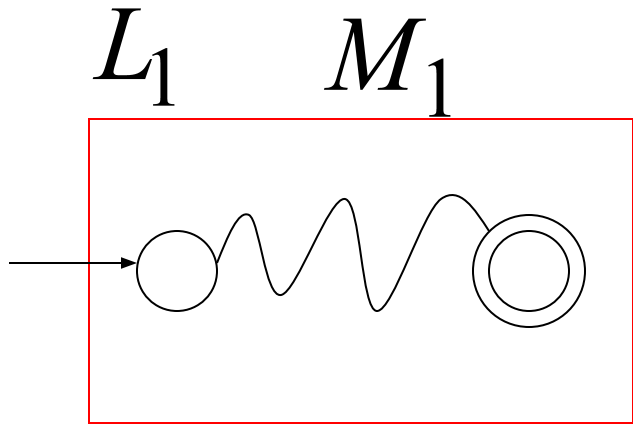
HOW?

Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$

# Reverse

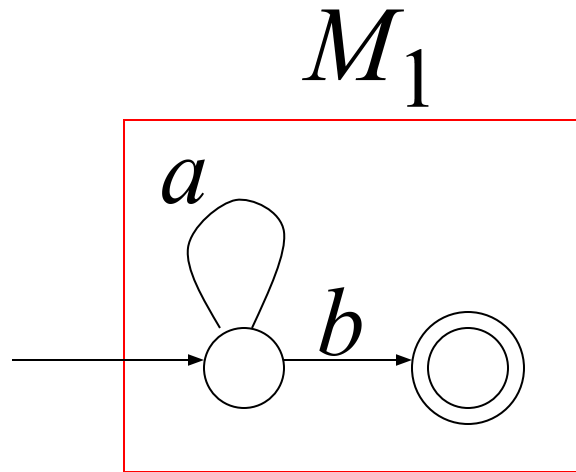
NFA for  $L_1^R$



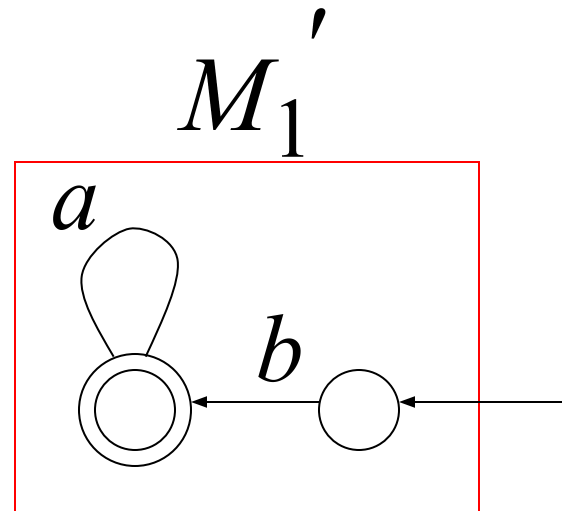
1. Reverse all transitions
2. Make initial state final state and vice versa

# Example

$$L_1 = \{a^n b\}$$



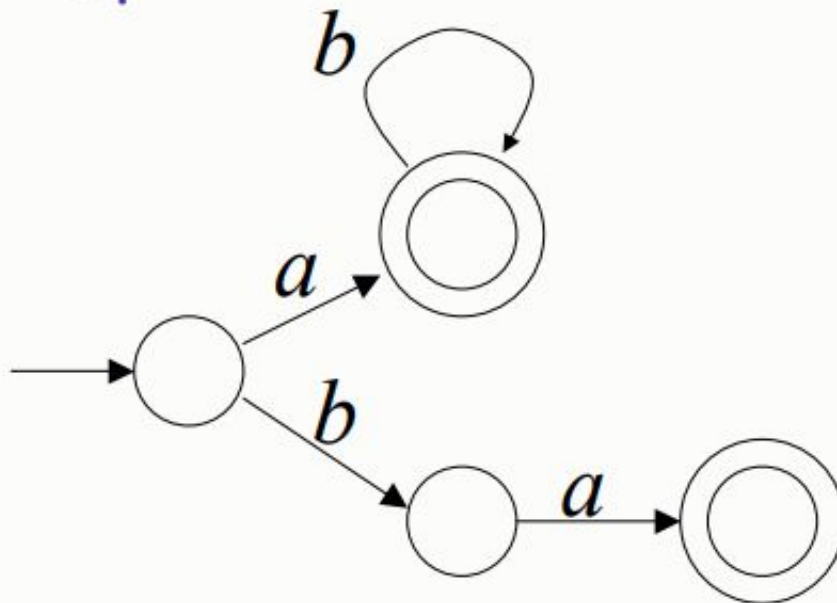
$$L_1^R = \{b a^n\}$$



# Proof

Since  $L$  is regular,  
there is NFA that accepts  $L$

Example:



$$L = ab^* + ba$$

Show Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

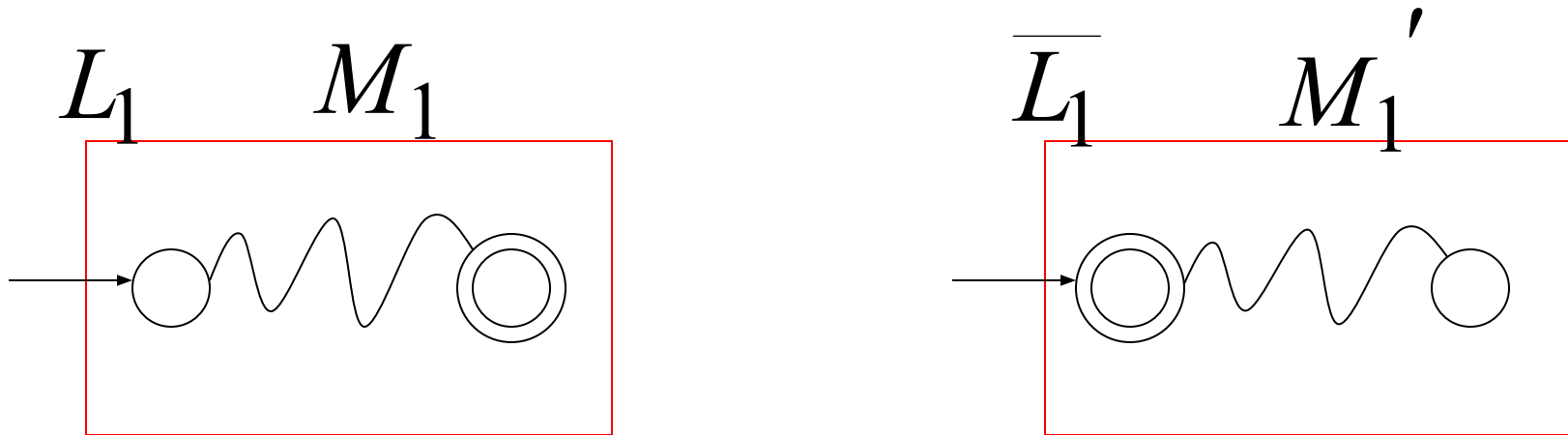
Reversal:  $L_1^R$

HOW?

Complement:  $\overline{L_1}$

Intersection:  $L_1 \cap L_2$

# Complement

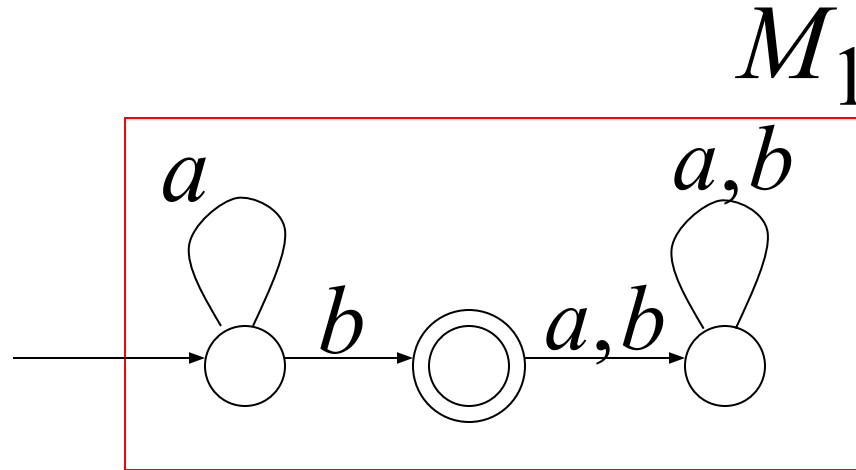


1. Take the **DFA** that accepts  $L_1$
2. Make final states non-final, and vice-versa

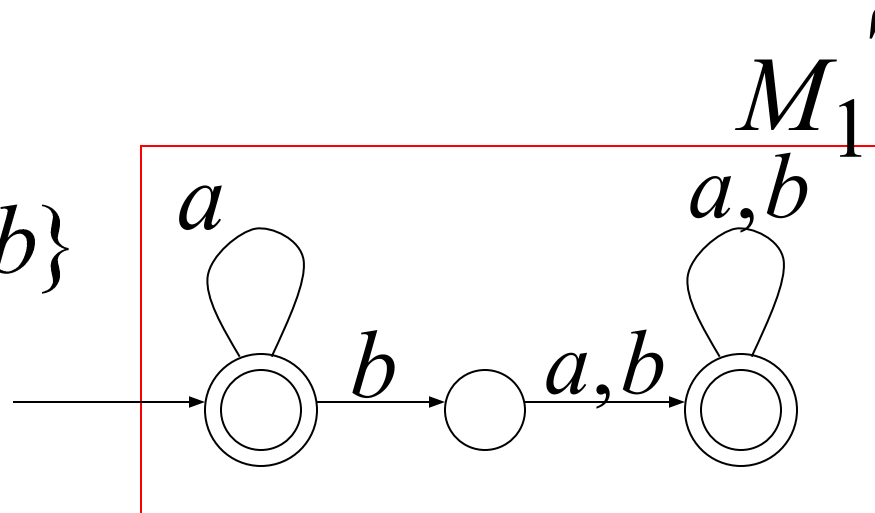


# Example

$$L_1 = \{a^n b\}$$



$$\overline{L_1} = \{a,b\}^* - \{a^n b\}$$



# Show Regular languages are **closed under**

Union:  $L_1 \cup L_2$

Concatenation:  $L_1 L_2$

Star:  $L_1^*$

Reversal:  $L_1^R$

Complement:  $\overline{L_1}$

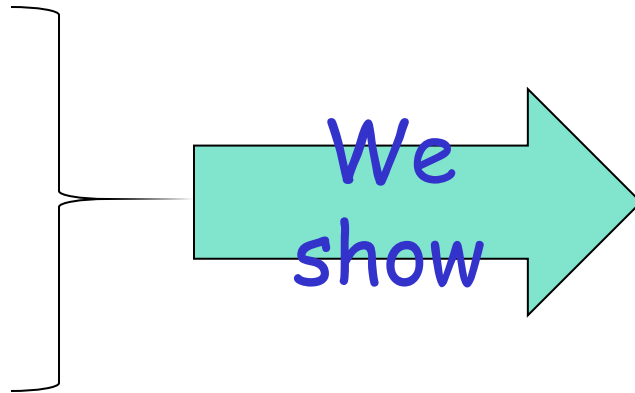
Intersection:  $L_1 \cap L_2$

HOW?

# Intersection

$L_1$  regular

$L_2$  regular



$L_1 \cap L_2$   
regular

# Proof for Intersection Closure

Machine  $M_1$

DFA for  $L_1$

Machine  $M_2$

DFA for  $L_2$

Construct a new DFA  $M$  that accepts  $L_1 \cap L_2$

$M$  simulates in parallel  $M_1$  and  $M_2$

$M$  simulates in parallel  $M_1$  and  $M_2$

$M$  accepts string  $w$  if and only if:

$M_1$  accepts string  $w$   
and  $M_2$  accepts string  $w$

$$L(M) = L(M_1) \cap L(M_2)$$

# Another Proof for Intersection Closure

DeMorgan's Law:  $L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$

$L_1, L_2$  regular

→  $\overline{L_1}, \overline{L_2}$  regular

→  $\overline{L_1} \cup \overline{L_2}$  regular

→  $\overline{\overline{L_1} \cup \overline{L_2}}$  regular

→  $L_1 \cap L_2$  regular

# Example

$$\begin{array}{l} L_1 = \{a^n b\} \text{ regular} \\ L_2 = \{ab, ba\} \text{ regular} \end{array} \bigg\} \xrightarrow{\text{yellow arrow}} L_1 \cap L_2 = \{ab\} \text{ regular}$$